TEXORA · LECTURE SHEET

Ultraviolet (UV) and Visible Spectroscopy

Beer–Lambert law with fully worked single-dye and two-dye calculations, the double-beam instrument, and Kubelka–Munk reflectance for dyed fabric.

TE-3105 · TEXTILE CHEMISTRY & COLORATION-II · TOPIC 11 · LEVEL 3 TERM 1

1. Light meeting molecules — the UV-Vis region

Ultraviolet (200–400 nm) and visible (400–700 nm) spectroscopy measures the electronic transitions of Topic 1: a solution absorbs photons whose energy matches its HOMO–LUMO and related gaps, and the absorption spectrum (absorbance versus wavelength) is a fingerprint of the chromophore. For dye chemistry the visible part does the daily work — λmax identification, strength comparison of dye lots, concentration measurement, and mixture analysis — while UV absorption tracks colourless auxiliaries, optical brighteners and degradation products.

2. The laws: Beer, Lambert, and additivity

$$\text{Transmittance } T = I/I_0 \quad | \quad \text{Absorbance } A = -\log_{10} T = \log_{10}(I_0/I) \quad | \quad \mathbf{A = \varepsilon \cdot c \cdot l}$$

Beer–Lambert law: absorbance is proportional to concentration c and path length l; ε (molar absorptivity, L mol^{-1} cm^{-1}) is the molecule's constant at a given wavelength and solvent. Consequences used daily: (1) A doubles when c doubles — calibration is a straight line through the origin; (2) absorbances of mixture components **add** at each wavelength, so two dyes are quantified from A at two wavelengths by simultaneous equations; (3) the law fails at high concentration (molecular association, scattering, stray light) — work in A ≈ 0.1–1.0 by dilution.

WORKED EXAMPLE 1 — CONCENTRATION FROM ABSORBANCE

A reactive dye solution in a 1 cm cell gives A = 0.840 at λmax. Its ε at λmax = 2.10 × 10^4 L mol^{-1} cm^{-1}.

c = A/(ε·l) = 0.840 / (2.10 × 10^4 × 1) = 4.00 × 10^{-5} mol/L.

With molar mass 600 g/mol: mass concentration = 4.00 × 10^{-5} × 600 = 0.0240 g/L = 24.0 mg/L. A 1:10 dilution was measured, so the original bath sample held 240 mg/L — the residual dye from which exhaustion is calculated in Experiment work.

WORKED EXAMPLE 2 — TWO-DYE MIXTURE

At λ_1 dye X absorbs strongly, dye Y weakly; at λ_2 the reverse (ε in L mol^{-1} cm^{-1}, l = 1 cm): εX_1 = 20,000, εY_1 = 2,000; εX_2 = 3,000, εY_2 = 18,000. Measured: A_1 = 0.660, A_2 = 0.570.

Equations: 20,000 cX + 2,000 cY = 0.660; 3,000 cX + 18,000 cY = 0.570.

Multiply the first by 9: 180,000 cX + 18,000 cY = 5.940. Subtract the second: 177,000 cX = 5.370 → cX = 3.03 × 10^{-5} mol/L.

Back-substitute: 2,000 cY = 0.660 − 20,000 × 3.03 × 10^{-5} = 0.660 − 0.607 = 0.0534 → cY = 2.67 × 10^{-5} mol/L. (Check in eq. 2: 0.091 + 0.481 = 0.572 ≈ 0.570 ✓, rounding.)

3. The instrument

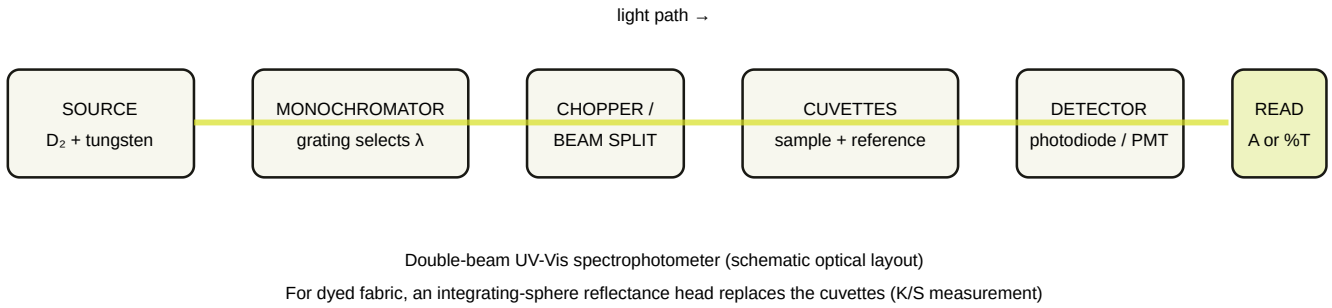


Fig. 1 — Optical layout of a double-beam UV-Vis spectrophotometer.

Deuterium lamp covers UV, tungsten covers visible; the grating monochromator selects wavelength; the beam alternates between sample and solvent reference so source drift cancels; the detector converts light to the A or %T reading. **Cuvettes:** quartz for UV (glass absorbs below ~350 nm), matched pairs, clean faces, no bubbles. **For solids** (dyed fabric) the reflectance spectrophotometer with integrating sphere measures R at each wavelength, converted by **Kubelka–Munk: K/S = (1−R)²/(2R)**, proportional to dye concentration on the fibre — the quantity used in recipe prediction (Topic 5) and in the migration index of Topic 7. Example: R = 0.20 → K/S = (0.80)²/0.40 = 1.60.

4. Analytical practice in the dyehouse laboratory

- **Dye strength comparison:** standard and delivery dissolved identically; ratio of A at λmax = relative strength (e.g. 0.92 → lot is 92% strength; recipes are corrected by 1/0.92).
- **Exhaustion measurement:** A of bath before and after dyeing (same dilution) gives % exhaustion directly, since A ∝ c.
- **Effluent colour** and residual dye are followed at λmax after calibration.

- **Deviations to suspect:** turbidity (scatter inflates A — filter or clarify), pH-sensitive dyes (buffer all solutions alike), and association at high c (dilute into the linear range).

5. Glossary — key terms

- **λmax:** wavelength of maximum absorbance; the analytical wavelength of choice.
- **Molar absorptivity ε:** absorbance of a 1 mol/L solution in a 1 cm cell.
- **Kubelka–Munk K/S:** function of reflectance proportional to colorant concentration on an opaque substrate.
- **Stray light:** off-wavelength radiation reaching the detector, flattening high absorbances.

6. Key points

- A = εcl, linear in the range A 0.1–1.0; mixtures add absorbances — two wavelengths solve two dyes.
- Solutions are measured in transmission; dyed textiles in reflectance as K/S.
- Every dyehouse number — strength, exhaustion, residual bath — traces back to this one law.

7. Summary

UV-Vis spectroscopy converts the colour concepts of Topic 1 into laboratory measurement. Beer–Lambert proportionality, additive absorbances and Kubelka–Munk reflectance together give the dye chemist concentration, mixture composition and on-fibre depth from a beam of light — closing the loop of this theory course.

8. Practice

1. A dye ($\varepsilon = 1.50 \times 10^4$, M = 500 g/mol) reads A = 0.750 in 1 cm. Find c in mol/L and mg/L.
2. Bath absorbance falls from 1.20 (before) to 0.30 (after), same dilution. Calculate % exhaustion. Why must the dilution be identical?
3. Fabric reflectance at λmax is 0.10. Calculate K/S and compare depth with the R = 0.20 fabric of the text.
4. A calibration bends downwards above A = 1.2. Give two instrumental/molecular reasons.

Answer key

1. c = 0.750/15,000 = 5.00×10^{-5} mol/L; × 500 g/mol = 0.0250 g/L = 25.0 mg/L.
2. Exhaustion = (1.20 − 0.30)/1.20 = 75%. A ∝ c only per unit dilution; different dilutions scale the readings differently and the ratio would be meaningless.

3. K/S = $(0.90)^2/(0.20)$ = 0.81/0.20 = 4.05, versus 1.60 — about 2.5× the dye concentration (K/S ratio 4.05/1.60 = 2.53), consistent with the darker fabric.
4. Stray light adds a constant signal that dominates when true transmitted light is tiny, capping apparent A; and at high concentration dye molecules associate (dimerise), changing ε — the solution no longer contains the species the law assumes.
